Expt. 3 γ-spec. (E) SCA

tions between modules. For this reason, essentially two standardized NIM systems have arisen: one designed for fast nanosecond pulses and the other for slower pulses. Within each system the electronic modules are compatible with each other; however, mixing the two without proper adaptation will invariably lead to problems. And, as already mentioned, special attention must be paid to how the interconnections are made in fast systems. This is one of the most important points to keep in mind when setting up nuclear electronics.

11.4 The Frequency Domain. Bandwidth

Our description and visualization of pulses so far has been in terms of its variation in time. A complete understanding of pulse electronics, especially pulse distortion, however, also requires viewing the pulse in terms of its frequency components. From Fourier analysis, it is well known that a pulse form can be decomposed into a superposition of many pure sinusoidal frequencies. Indeed, if we have a pulse whose shape in time is represented by the function $f(t)$, where t is time, then it may be decomposed as

$$f(t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} g(\omega) \exp(i\omega t)\, d\omega \,, \tag{11.1}$$

where $g(\omega)$ is the *Fourier transform* or frequency spectrum of the pulse. Inverting (11.1) gives

$$g(\omega) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(t) \exp(-i\omega t)\, dt \,. \tag{11.2}$$

As a representative example, consider an ideal rectangular pulse of width T shown in Fig. 11.3. For simplicity, we have taken the pulse to be centered at $t = 0$. Thus,

$$f(t) = \begin{cases} A & |t| < T/2 \\ 0 & |t| > T/2 \,. \end{cases} \tag{11.3}$$

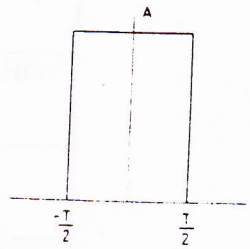
If we Fourier analyze this function, we find the spectrum

$$g(\omega) = \frac{1}{\sqrt{2\pi}} \int A \exp(-i\omega t)\, dt = \frac{AT}{\sqrt{2\pi}} \frac{\sin(\omega T/2)}{(\omega T/2)} \,, \tag{11.4}$$

which is plotted in Fig. 11.3 as a function of frequency $f = \omega/2\pi$. As can be seen, $f(t)$ contains a continuous spectrum of frequency components from 0 to ∞, arranged in a band structure reminiscent of optical diffraction. The energy or power contained in each frequency component is the square of $g(\omega)$:

$$E(\omega) = |g(\omega)|^2 \,. \tag{11.5}$$

The negative frequencies are, of course, purely imaginary.



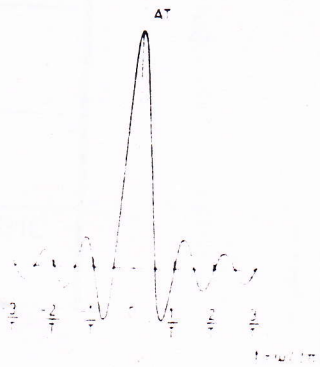


Fig. 11.3. Fourier spectrum of a rectangular pulse

Exp. 3

γ- Spectrometer (SCA) (E)

Preamplifier and Cable Driver circuit is available in **NETS0M** and **NETS0MHC** models. The Anode signal of voltage divider circuit, de-coupled through capacitor is fed to the input stage of the low noise preamplifier circuitry. The necessary low voltage ±15VDC power is fed through 3 pin connector.

Cable driver circuit at output stage of **NETS0M** and **NETS0MHC** is capable of driving signal through a standard RG 58/62 type, coaxial cable up to a maximum length of 50 metres. The output signal is on BNC connector marked with OUTPUT.

1.3 Specifications

Technical Specifications

Specification	Values	Available in Models
Voltage Divider	10 or 12 stage with R=510 ohms	All models
Preamplifier	2 stage transistor amplifier with emitter follower	NETS0M & NETS0MHC
Signal output	Output pulse of Voltage Divider stage	NETS0VD
Signal output	Pulse output with sharp rise time and 50 µSec fall time	NETS0M & NETS0MHC
HV supply	Screw driver adjustment for setting HV between +100VDC to +1000VDC	NETS0MHC
HV monitor	0 to 5V output proportional to HV.	NETS0MHC
Power Supply	±12V to ±15VDC @150 mA.	NETS0M & NETS0MHC

General

Mechanical	:	Base Stand or In-line package
Electrical	:	±12 V to ±15VDC supply for Preamplifier
Weight	:	Approximately 500 gms

1.4 Controls / Indicators / Connectors

Model NETS0VD :

HV	-	Mini UHF connector for high voltage supply.
OUTPUT or O/P	-	BNC connector for signal output

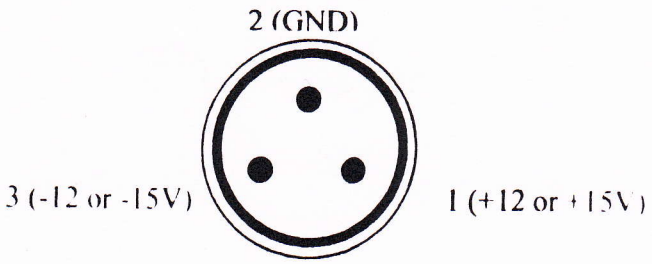
Model NETS0M :

HV	-	Mini UHF connector for high voltage supply.
OUTPUT or O/P	-	BNC connector for signal output
± 12 V / ±15V	-	3 pin connector for low voltage supply

Model NETS0MHC :

OUTPUT or O/P	-	BNC connector for signal output
± 12 V / ±15V	-	3 pin connector for low voltage supply
HV ADJ	-	Screw driver adjustment for setting HV between +100V to +1000V
HV and GND	-	Test points to monitor HV value. (0 to 5VDC correspond to 0 to 1000VDC)

3 Pin Male Connector for power input as seen from solder side is shown below.



2.0 INSTALLATION

2.1 Unpacking and Handling

Carefully unpack NETS0. Please notify the factory or nearest sales office immediately in case of any physical damage notified.

In case of any physical damage during transportation, **DO NOT APPLY POWER** to NETS0 and notify the factory or nearest sales office immediately.

2.2 Power Requirement

NETS0 requires two DC supplies, one for PMT and another for preamplifier circuitry. Both power supplies can be connected through the 3 pin connectors mounted on the NETS0 housing. The high voltage supply should be set to the value recommended by the manufacturer of the detector. The typical value of HV is around +750 Volts.

3.0 OPERATION

Following procedure should be adopted for trouble free operation of NETS0.

1. Connect the HV supply cable to the Mini UHF connector marked as HV. (Not applicable for NETS0MHC model.)

2. Connect ±15V supply to 3 pin connector. (Not applicable for NETS0VD model). Care should be taken to connect proper polarity. Reverse polarity will damage the sensitive circuitry.

3. Plug the NaI(Tl) detector PMT in the PMT base.

4. Apply HV and Preamp power as mentioned above.

5. Expose the Detector Window surface to a radio-active source.

6. Check output of Preamp on oscilloscope. Pulses with sharp rise time (few nsec) and fall time (few μsec) will be seen on the scope.

7. Connect this output to spectroscopy amplifier input to check for the shaped output.

4.0 MAINTENANCE

NETS0, PMT based Preamplifier has been carefully designed for trouble free service. It is not possible to repair it at site should the need arises. It is recommended that the unit should be returned to factory for necessary repairs, with prior intimation.